# MAHARASTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI - 51

Government Polytechnic Building, 49, Kherwadi, Bandra (E), Mumbai - 400 051.

## Photocopy Marks Summary :: Winter - 2025

Enrollment No. : **23111580076**

Br. & Sem. : **CO - 5**

Seat No. : **121165**

Subject Code : **315323**

Student Name : **SAHU YASH YOGESH**

Subject Name : **SOFTWARE ENGINEERING**

| Sr. No. | Q.No. | Main Marks | After Photocopy Marks |
|---|---|---|---|
| 1 | 01a | 1.5 | 1.5 |
| 2 | 01b | 1.5 | 1.5 |
| 3 | 01c | 2 | 2 |
| 4 | 01d | 2 | 2 |
| 5 | 01e | 1* | 1* |
| 6 | 01f | 2 | 2 |
| 7 | 01g | NA* | NA* |
| 8 | 02a | NA | NA |
| 9 | 02b | NA* | NA* |
| 10 | 02c | 3 | 3 |
| 11 | 02d | 2.5 | 2.5 |
| 12 | 03a | NA* | NA* |
| 13 | 03b | 3 | 3 |
| 14 | 03c | 3 | 3 |
| 15 | 03d | 3 | 3 |
| 16 | 04a | 3 | 3 |
| 17 | 04b | NA* | NA* |
| 18 | 04c | 3 | 3 |
| 19 | 04d | 0.5* | 0.5* |
| 20 | 04e | 3 | 3 |
| 21 | 05a | 3 | 3 |
| 22 | 05b | 3 | 3 |
| 23 | 05c | NA* | NA* |
| 24 | 06a | 3.5 | 3.5 |
| 25 | 06b | 4 | 4 |
| 26 | 06c | NA* | NA* |
| | Total | **46** | **46** |

NA : Indicates Question not attempted.

Note : * - indicates Extra question, not considered in Total.

# MAHARASHTRA STATE
# BOARD OF TECHNICAL EDUCATION, MUMBAI

2425

(ANSWER BOOK)

1001230

To be filled

Subject :- Software Engineering

Subject Code :- 3 1 5 3 2 3

Section - I / II

Course and Year/Sem :- Cosn

Date of Exam :- 1 D Y M 2 0 2 5

| Main Ans. Book | No. of Supplements | Total |
|---|---|---|
| 1 | 0 | 1 |

Please read instructions to examinees overleaf

Sr. No. 30534794

Examination Seat No.

1 2 1 1 6 5

Exam. Center Code: 0 5 6 8

Institute Code: 0 5 6 8

Signature of block Superviser / Invigilator

14/4/25

| Q. No. | a | b | c | d | e | f | g | h | i | j | k | l | m | n | Total | Q.C. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | |
| 2 | | | | | | | | | | | | | | | | |
| 3 | | | | | | | | | | | | | | | | |
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| 6 | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | |
| Sign and Name of Examiner | | | | | | | | | | | | | Total | | | |
| 1 | | | | | | | | | | | | | | | | |
| 2 | | | | | | | | | | | | | | | | |
| 3 | | | | | | | | | | | | | | | | |
| 4 | | | | | | | | | | | | | | | | |
| 5 | | | | | | | | | | | | | | | | |
| 6 | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | |
| Sign and Name of Moderator | | | | | | | | | | | | | Total | | | |

## INSTRUCTIONS TO EXAMINEES

**Following instructions are mandatory for all examinees in addition to the instructions on their hall ticket for the examination.**

1. The examinee shall check the answer book issued to him/her for ensuring that there are no loose pages, improperly printed pages, missing pages or torn pages etc. and if any such defect is noticed, the answer book shall be replaced by reporting to the block invigilator before commencing writing anything on or inside it.
2. Examinees are not permitted to leave examination hall in the initial 30 minutes and last 10 minutes of the examination duration. If examinee is leaving the examination hall before the end of examination, he/she shall submit the question paper along with the answer book.
3. A warning bell will be sounded 10 minutes before the close of the examination. Examinees shall tie the attachments if any to the answer book and be ready to hand over it to the invigilator at the ringing of the final bell announcing the end of the examination.
4. The examinee shall furnish all the requisite information expected to be filled by examinee on the face sheet of the answer book properly before commencing to write the answers.
5. Use only blue or black ink pen to write answers. If there is change in ink, it shall be attested by the supervisor/ invigilator.
6. If the "examinee wishes to cancel the performance of the current paper after writing the answers in the answer book, the same can be done by scratching all the written answers and writing the remark as under;
   - "I have scratched all my answers and wish that the answers in this answer book shall not be assessed" at the end of the cancelled answers.
   - The block invigilator shall sign the answer sheet by putting the remark "the scratching of answers and the examinee's remark are seen by me."
   - In case this Instruction is not followed, then the scratched answers shall be assessed at the Regional Assessment Center.
7. **Important note for the Examinees of PH1J and PH2J: In case, multiple answer options are observed for the same sub question of question No. 3, the option appearing first in the answer book shall be treated as answer and assessed accordingly. Hence the examinee shall retain only one answer option for each sub question of question No. 3 by striking off remaining options for the same sub question.**
8. Exchange of answer books, question papers, calculators, drawing instruments etc. with other examinees in the examination block is strictly prohibited. Such an act shall amount to malpractice.
9. Disclosure of identity of the examinee by writing name / examination seat number / enrollment number / appeal to examiner or writing objectionable matter inside the answer book is strictly prohibited. Such an act amounts to malpractice.
10. Do not tear off any page from the answer book it amounts to punishment. If tear off page is by mistake get it certified from the supervisor/invigilator in the answer book.
11. The examinee shall behave properly before, during or after the examination to maintain the decorum of the examination. The violation of this instruction shall amount to malpractice.
12. The examinee found guilty of malpractice shall be liable for suitable punitive actions as per the provisions of the examination regulations of MSBTE.
13. Possession of any arms, weapons, etc. in the examination hall or at the examination center by the examinee is strictly prohibited.
14. Start each answer on a fresh page and write question number at the beginning of each answer. Do not write anything in the margin of answer book.
15. The examinee shall use separate answer book for each section, where there are sections in question paper.
16. Do not leave blank page/s between the answers. If a page is left blank inadvertently, write "Please Turn Over (PTO)." Answers written beyond a blank page may not be assessed.
17. Examinees shall not write anything on the question paper of the examination except for his / her seat number. Such writings on question paper or scribbling amount to malpractice and is liable for punishment.

Q1.

c) Ans-

Principles of Deployment

(i) Manage customer expectations

2 01c

Before satisfying all customer needs, the customer may not be fulfill to all his requirements, because of last-minute changes. A developer must have the ability to manage customer requirements and satisfy the customer.

(ii) Don't give customer buggy/defective software

In incremental type of software process sometimes in initial increments the developer dates deliver some buggy software and promises to fix it in next increments.

(d) Ans- (i) Abstraction

It is the process of hiding unnecessary information and showing useful and essential information to the user to reduce complexity.

2 01d

(ii) Modularity:

It is the process of dividing a large software into smaller manageable parts, often known as modules. It helps for maintanence of the program/software.

Q. 1)

Ans- (i) Large main deliverables are divided into small chunks, helps for easy maintainence and management of software.

2 01f

(ii) Early access to problem and risk, as they are divided in parts, suitable for component-level risk analysis.

(e) Ans-

A risk is an ~~exp~~ expectation to loss, losses due to lack of information, control or time.

1 01e

(b) Ans- Software engineering is the fundamental technique of using software engineering principle and practices for the development of efficient software.

1.5 01b

(i) Application Software

(ii) System Software.

(c) 1.5 01a

(i) Less Complex (iii) User friendly

(ii) Durable (iv) Lightweight.

Q2.

(d) Ans-

(i) Compartmentalization:

The Software project must be compartmentalized into small parts that helps for easy managing and scheduling. Small parts are basically called tasks and activities.

(ii) Interdependency:

2.5 02d interdependency of each compartmentalized part should be determined. ~~Each~~ Some task can be performed in serial while others can occur parallel. Activities occurs independently.

(iii) Time allocation:

Each task must be assigned a ~~ded~~ fixed time period, to complete and deliver it on time. Work units, schedules, resource allocation are done.

~~(iv) Effort validation~~

(iv) Defined responsibilities:

Each task must be associated to a specific team member. This will help in faster execution

Q2 c)

Ans-

| Element | name | Task performe |
|---|---|---|
| → | Data flow | Used to connect processes, to source or sink, the arrow head shows the flow of data |
| ○ | process | A specify task performed |
| ↓ ↑ | Data repository | A place where large data is stored and can be use later. |
| □ | Source or Sink | Source out to process and sink in |

3 02c

Q.

(c) Symbol:

(i) A circle is usually used to represent the proces, that Operforms ~~to a~~ ~~[illegible]~~ Some task, there can multiple processes.

(ii) A arrow represents the data flow that connects different process. The head of arrow decides the flow of data.

(iii) A parallel lines represent the data repository a place where data is stored and can be used later.

(iv) A rectangular box represents a source act and sink in to process.

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Q3.
d) Ans-

Consider

P for permanent
NP for Not permanent
A for Above 50
B for 50 and below 50
F for full insurance
PF for Partial insurance
NF for No insurance

| Criteria | Case 1 | Case 2 | Case 3 | Case 4 |
|---|---|---|---|---|
| Employee Status | P | P | NP | NP |
| Age | A | B | A | B |
| Output (if he gets insurance) | F | PF | NF | NF |

3 03d

Case 1 if permanent and above 50
so full insurance

Case 2 if Permanent and below 50
partial insurance

Case 3 if not permanent and above 50
no insurance

Case 4 if not permanent and below
50 still no insurance.

Q3 (c)

Ans-

| White box testing | Black box testing |
|---|---|
| (i) The tester ~~doesn't~~ does need to have the knowledge about code and program. | (i) The tester doesn't need to have the knowledge about code and program |
| (ii) It tests the structure of the item. | (ii) It tests the functionality of software. |
| (iii) Also known as structural, clear-box testing. | (iii) Also known as functional, closed box testing. |
| (iv) Best suited for low-level testing like unit testing. | (iv) Best suited for high level like system testing. |

3 03c

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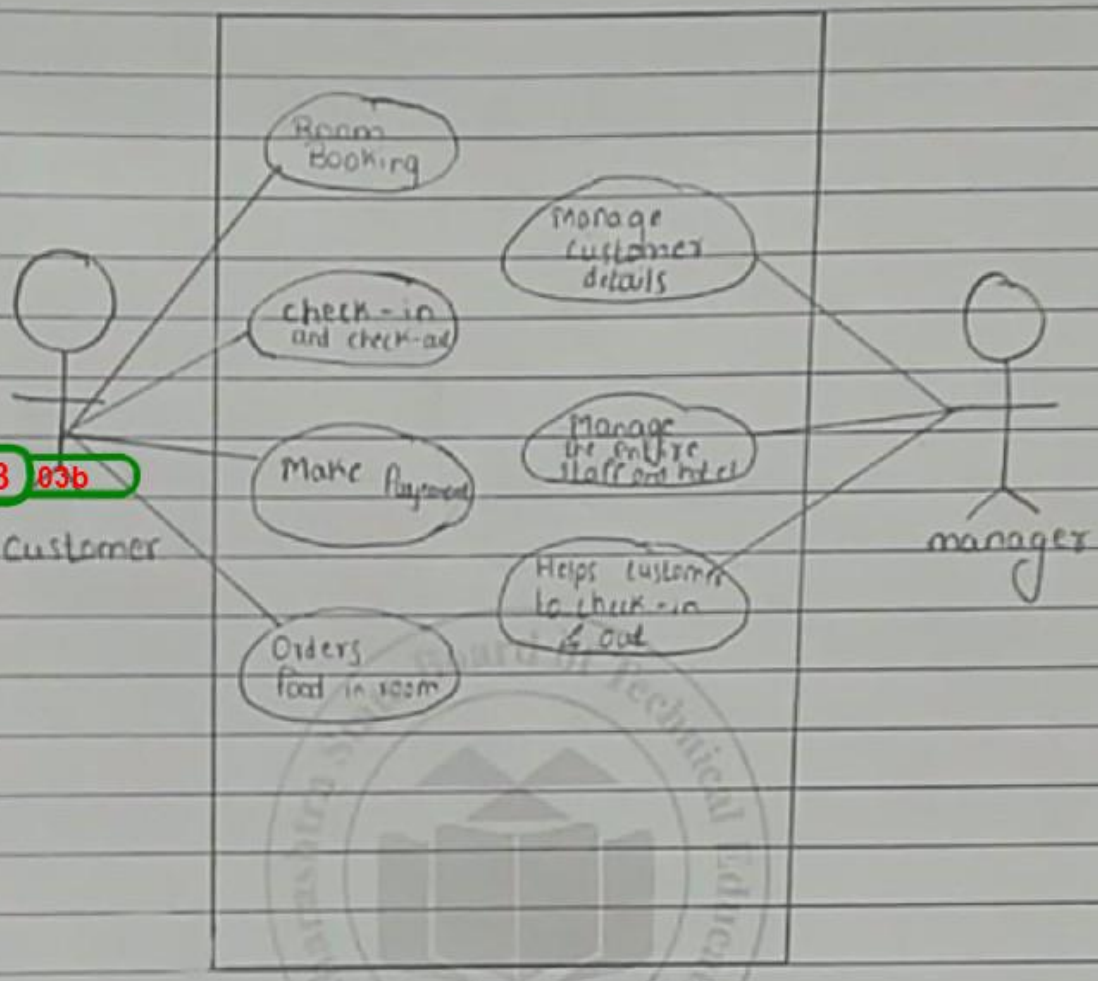
Q3.

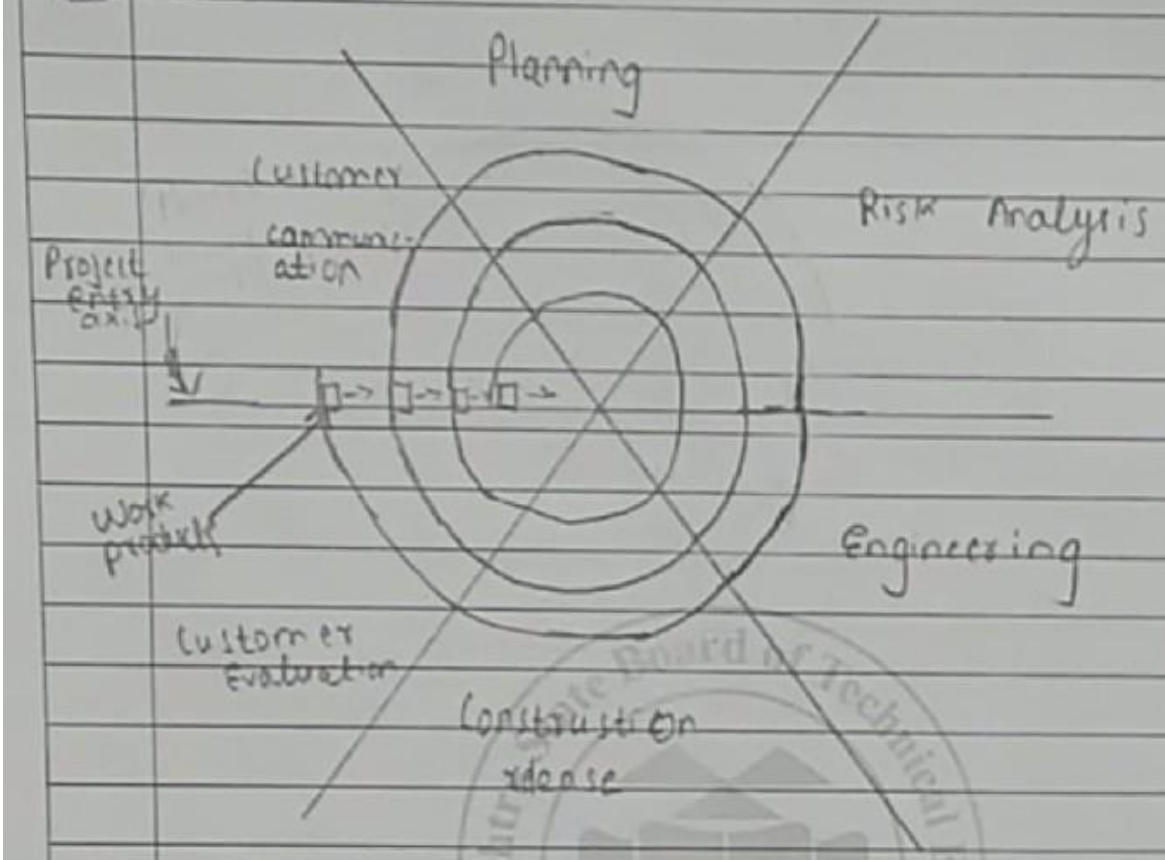
b) Ans-




3 03b

use case diagram of hotel management

Q4.
a) Ans-




3 04a

Spiral model

(i) Communication:
This phase e involves heavy communication with customer to known the requirements. This will help to have a better understanding to the project.

(ii) Planning:
Resource gathering is done and technical task, work unit and schedules are being assigned.

) Risk Analysis:
Identification of possible risks are done here to be ready if any of them occur in future with back-up strategies. This helps the organization a lot in terms of technical and business loss both.

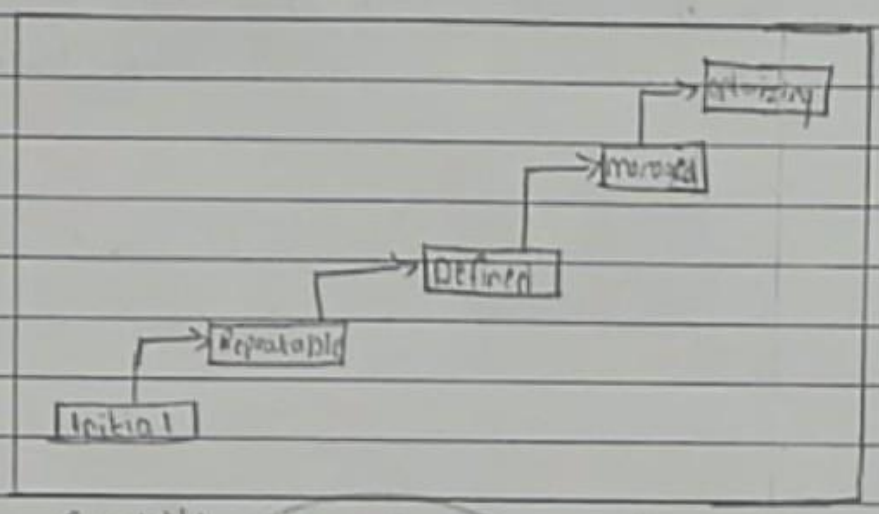
(iv) Engineering:
The actual development starts here. The main work products of this phase are design, code and other related documents.

(v) Construction and test release
The complete software package is prepared user manual and guide along with source code and other stuff from engineering phase and is released.

(vi) Customer evaluation
The customer evaluates the software and gives feedback and suggest changes if needed. If there would be any changes then again the steps are followed and changes are made in next spiral.

Q4.
(c) Ans.

(i) Diagram.




Capability Maturity model Integration (CMMI).

(ii) Description of each level

3 04a

(1) Initial : Level 1:
The software process is characterized as ad hoc and occasionally even chaotic.

(2) Repeatable : Level 2:
The cost estimation, process tracking, scheduling and other related process stuff are done.

(3) Defined : Level 3:
The requirements of customer and other related thing are performed.

(4) Managed : Level 4:
The software process is managed. Errors being detected so it no not affect in future.

Q4)

d) Optimizing : Level 5:

The entire thing is always in way to be more optimized and improved by ~~continuous~~ continuous feedback and exploring new ideas and technologies.

0.5 048

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Q4)

e) Ans- (i) Risk mitigation, monitoring and management plan (RMMM) is a plan to avoid and prevent risk and manage the system to keep ~~bugg~~ error-free and risk-free.

(ii) Management can be incorporated in the 3 04e ~~ctive~~ system or can be maintained as a seperate in RMMM.

(iii) The management ~~&~~ spectrum describes that if avoidance and mitigation are not enough, then a management plan is carried to identify early risks.

(iv) This helps in having a back-up ~~stategeries~~ ~~start~~ strategies which will eventually ~~a~~ lead to less impact.

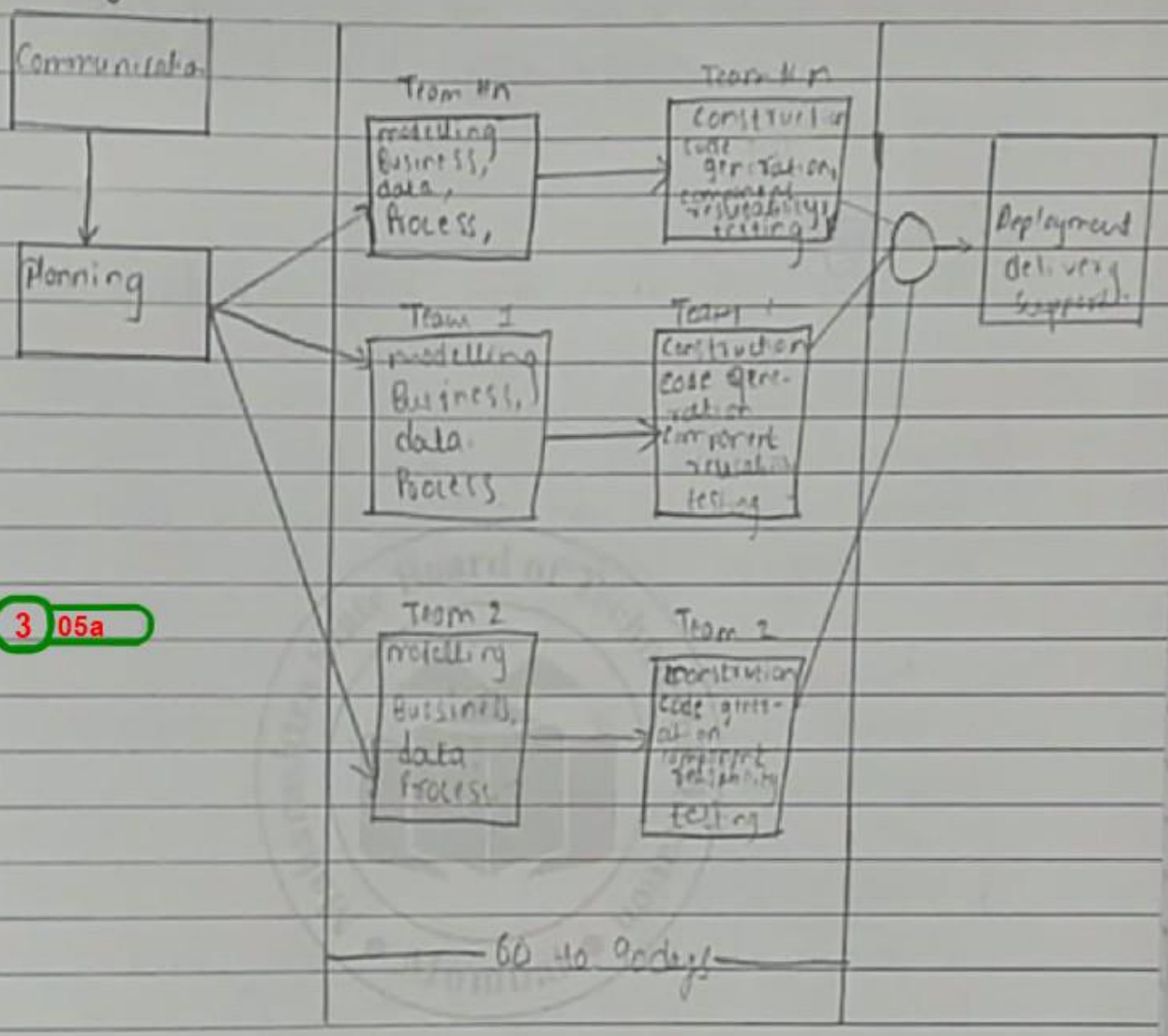
(v) The only motive behind this is to keep the system risk-free and also be aware of the possible risk in future.

(vi) Management is important task in every software development as it saves organization or companies from severe losses.

(vii) It also keeps in mind that the previous steps are applied properly to have a efficient software.

Q5.

a) Ans (i) Diagram




3 05a

RAD model.

(ii) Phases of RAD

(1) Communication:

Information gathering and requirements are discussed with the customer for a better understanding.

(2) ~~(ii)~~ Planning:

Requirement gathering and work unit, Schedule assignment is done.

(3)
(iii) Modelling :
Here it divided into multiple teams but the main purpose is to develop model for better understanding but in a faster way.

(i) Business model :
Business objectives are being modelled to understand business goal better.

(ii) data model :
~~Th~~ The data objects are converted into ~~Proje~~ Process related stuff.

(iii) Process model :
It's converted into business objectives and related stuff.

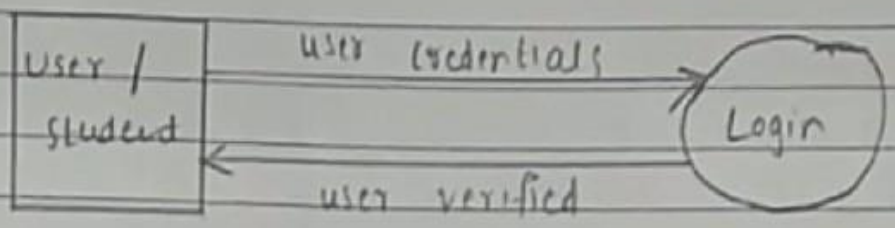
(4) Construction :
(i) Things like code generation and Component reusability is done for faster development.

(ii) Testing is also carried out in this stage.

(5) Deployment :
Lastly, all the work is integrated and delivered and feedback is taken for improvements.

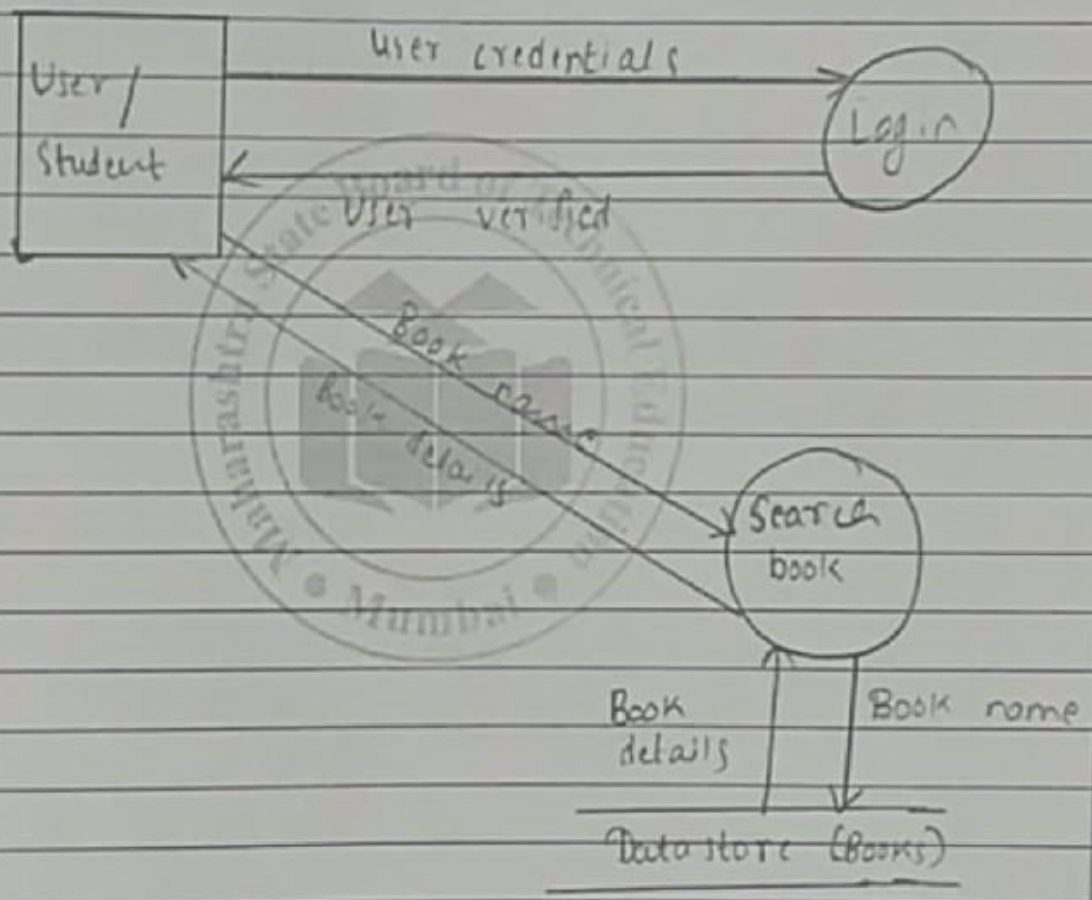
Q5)

b) Ans-




Level 0

3 05b




Level 1

Q6)

(b) Ans- KLOC = 400

Organic mode

$a = 2.4,\ b = 1.05,\ c = 2.5,\ d = 0.38$

Effort (E) = $a(KLOC)^b$ {a and b are constants}

$= 2.4(400)^{1.05}$

$= 1295$ staff

Development Time (TDEV) = $c(E)^d$ {c and d constants}

4 06b

$= 2.5(1295)^{0.38}$

$= 38.07$

$\approx 38$ hrs

Semidetached mode, $a = 3.0,\ b = \cancel{0.35}\ 1.12,\ c = 2.5,\ d = 0.35$

Effort (E) = $a(KLOC)^b$

$= 3.0(KLOC)^{1.12}$

$= 2462.7$ staff $= 2462$ staff

$\approx 2463$ staff

Development Time (TDEV) = $c(E)^d$

$= 2.5(2462)^{0.35}$

$= 38.46$

$\approx 38$ hrs

Q5

(b) Embedded mode

$a = 3.6$, $b = 1.20$, $c = 2.5$, $d = 0.32$ (constant)

Effort (E) $= a(KLOC)^b$

$= 3.6(4.00)^{1.20}$

$= 4772$ Staff.

Time for Development (TDEV) $= c(E)^d$

$= 2.5(4772)^{0.32}$

$= 37.5$

$\approx 38$ hrs

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(06)

(b) Ans-

(1) Listen:

(i) Pay attention towards what user is saying instead of framing answers for them.

(ii) Clear if anything is misunderstood but avoid constant interruptions.

3.5 06a

(2) Prepare before you communicate:

(i) Always have a detailed information about the communication.

(ii) If you got to delivery some thing prepare accordingly.

(3) Some one should facilitate the meeting.

(i) There is a facilitor who should manage the communication & see if all aspects are discussed properly.

(4) Take notes and document points:

(i) Important notes and things should be write down and document properly for further use.

(5) ~~Someone should~~ Record the minutes of meeting:

(i) Somebhe participating in the meeting should have a record of the minutes of meeting. This is important while having communication.

Q

1(a) (vi) Draw Diagram if any thing is uncleared:

(i) If anything is misunderstood in verbal form draw diagram or visual representation for better understanding to customer and other people to avoid confusion.

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Rough work

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Yes ~~Atomical~~

Liliu

Prepare in advance ~~for~~ to communicate

Someone should facilate the meeting

Take Note and document point